

IF SOCIAL IMPACT PRIZE · PROJECT DOSSIER

FloodSight

Street-level flood early warning for Lagos, Nigeria — built entirely on free, open data, delivered by SMS.

Lagos floods every rainy season. Warnings exist — but they arrive as state-wide announcements, not as a message that tells a family on a specific street to move their belongings tonight. FloodSight closes that last mile: satellites and open data score flood risk for every 200-metre cell of the city, a live rainfall and tide engine raises Watch and Warning levels, and residents receive plain-language SMS alerts on any phone — no smartphone, no app, no data plan.

15 LGAs

24,933 cells scored at 200 m — one flood-risk map for the whole city

14 / 18

documented Lagos floods (2011–2024) detected in independent back-testing

~ €0

software cost: 100% open data, open source, free-tier infrastructure

1 SMS

is all it takes to be warned — or to report back what really happened

CONTRIBUTES TO THE UN SUSTAINABLE DEVELOPMENT GOALS

SDG 11 Sustainable cities

SDG 13 Climate action

SDG 3 Health & well-being

SDG 1 No poverty

Rankine Innovation Lab

Applicant · Lagos, Nigeria

Live system: floodsight-starter.onrender.com

Project page: www.rankineinnovationlab.com/floodsight.html

Every number in this dossier is reproducible from the project's open codebase and validation logs.

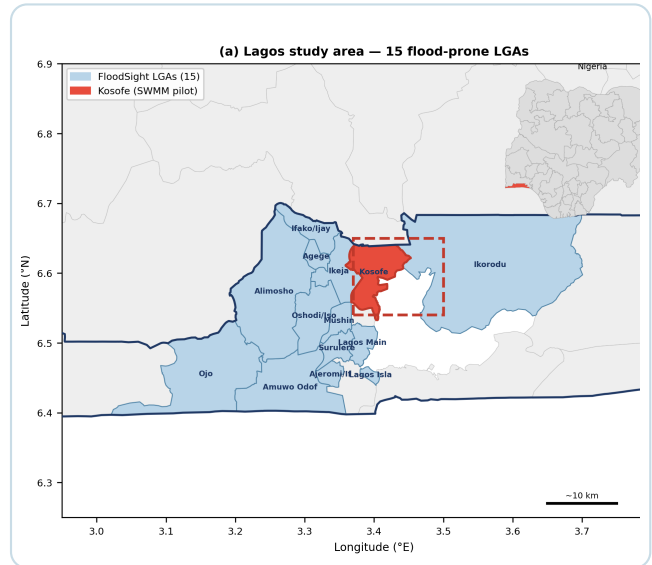
A megacity that floods every year — and a warning gap

Lagos is a low-lying coastal megacity of more than 15 million people. Between 2011 and 2024 it recorded major flood events in nearly every rainy season — the 2011 floods alone displaced thousands (NEMA), the June 2020 event swept a child away in Orile-Agege, and July 2021 brought tidal flooding to Lekki. Flooding is not a surprise here; it is a season.

Yet the warning chain is broken at the last mile.

Meteorological bulletins are issued state-wide, in technical language, through channels — TV, websites, press releases — that rarely reach the informal settlements and low-income districts that flood first and recover slowest. A resident of Kosofe or Ajegunle typically learns of a flood when water enters the house.

The gap is not data. Elevation, land cover, population, rainfall and radar imagery of Lagos are all freely available from public satellite programmes. What was missing is the design work that turns that data into one clear sentence, delivered to the right street, at the right time, on the phones people actually own.



Coverage: 15 flood-prone Local Government Areas of Lagos State (blue), from Ojo in the west to Ikorodu in the east. Red: Kosofe, the drainage-model pilot.

Designed for the people warnings usually miss

Residents of flood-prone streets

Plain-language Watch / Warning SMS for their exact location — on any phone. Subscribing takes one form or one text; leaving takes one word: STOP.

Community reporters

After an event, anyone can text FLOOD or NOFLOOD (with a depth, e.g. FLOOD 0.5) and their report becomes ground truth that retrains the model.

Emergency agencies (SEMA / LASEMA)

An operator dashboard shows live alert levels, street-level risk, subscriber coverage and dispatch history — evidence for pre-positioning response.

DESIGN PRINCIPLES

SMS-first, smartphone-optional

One message = one segment = one clear action

Calm language, never panic

Opt-out in every single message

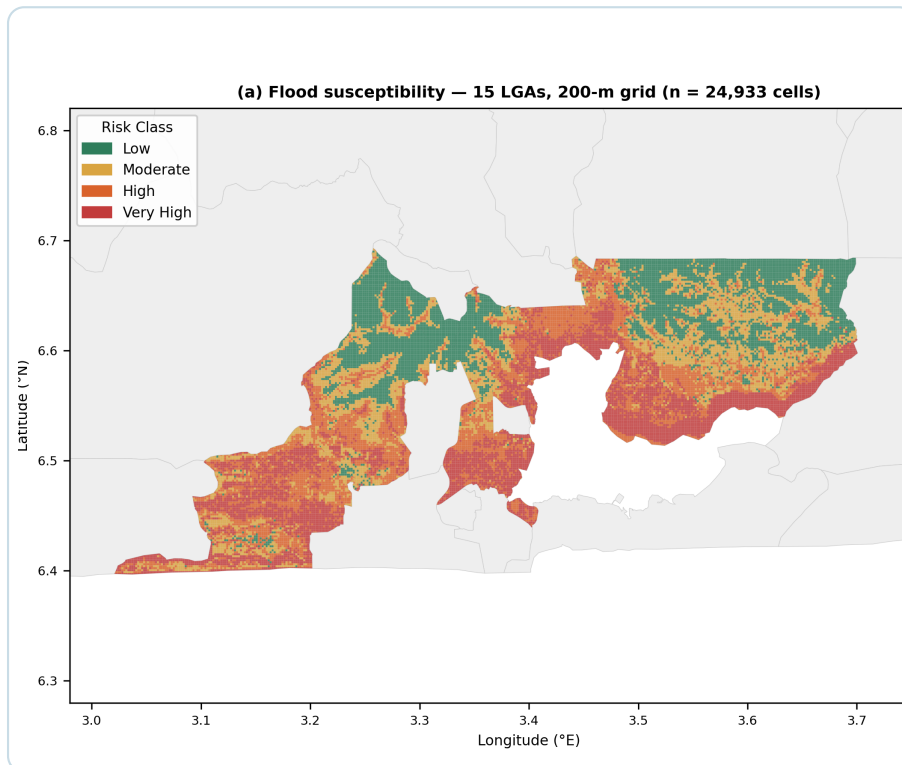
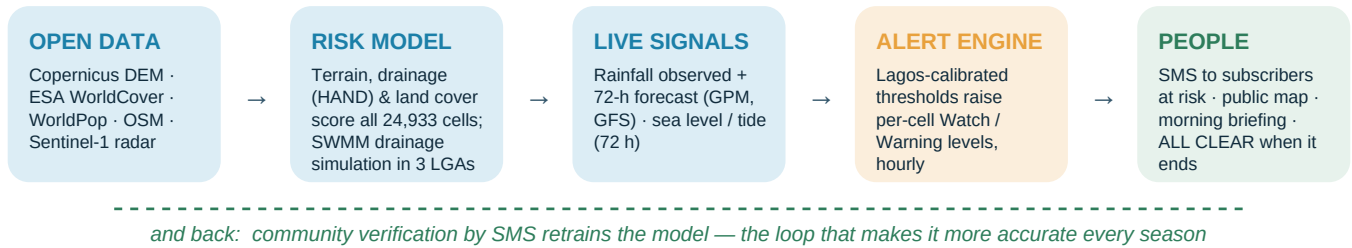
An explicit ALL CLEAR, not a fade-out

THIRTEEN YEARS OF DOCUMENTED LAGOS FLOODS — THE EVENTS FLOODSIGHT IS TESTED AGAINST



18 events, sourced from NEMA, LASEMA and FloodList situation reports — replayed against the alert engine as an independent back-test (see page 5).

Free satellites in, one clear sentence out



The city-wide flood susceptibility map: 24,933 cells at 200 m, classified Low to Very High from elevation, slope, drainage, land cover, distance to water and population — computed entirely from free satellite data.

Why open data matters here

Commercial flood models price out the cities that need them most. FloodSight's entire stack — elevation, land cover, population, rainfall, radar and tide — is free and global. The result: the software cost of protecting a megacity is effectively zero, and the same pipeline can be retargeted to Accra, Abidjan or Freetown by changing one configuration file.

Physics + learning, honestly labelled

A transparent, explainable susceptibility model is the backbone. On top of it, a hydraulic drainage simulation (SWMM, 6,340 modelled junctions across 3 LGAs) and a machine-learning depth model trained on 616 satellite-radar-labelled flood observations add street-level detail. Every training row carries a provenance label — observed vs. simulated — and the public API reports model accuracy only on real observations. No black box, no inflated claims.

Design here is not decoration. It is choosing the 160 characters that stand between a family and a metre of water.

THE OPEN-DATA STACK (ALL FREE, ALL GLOBAL)

Copernicus DEM

ESA WorldCover 10 m

WorldPop

OpenStreetMap

CHIRPS rainfall

GPM IMERG satellite rain

Sentinel-1 SAR

NOAA GFS forecast

Open-Meteo Marine (tide)

Four text messages that carry a whole service

DAY 1 · 08:00

FloodSight: Flood Watch for Kosofe. Heavy rain expected - avoid flood-prone streets. Map: rankineinnovationlab.com/floodsight Reply STOP to opt out

DAY 1 · 14:00

FloodSight ALERT: Flood Warning for Kosofe. Avoid low roads. Move to higher ground if needed. Reply STOP to opt out

RESIDENT REPLIES · DAY 2

FLOOD 0.5

DAY 2 · 09:00

FloodSight: All clear for Kosofe. Flood alert has ended. Stay careful near drains and canals. Reply STOP to opt out

● Every message is engineered

Alerts fit a single GSM-7 SMS segment (≤ 160 characters) so they cost one message on every Nigerian network, always name the recipient's area, always carry the opt-out — verified by automated tests on every release.

● The conversation goes both ways

Replying STOP unsubscribes instantly; START re-joins. Texting FLOOD 0.5 or NOFLOOD files a geo-referenced field report against that day's alerts — residents become the sensor network that audits and retrains the system.

● An honest ending

When conditions pass, subscribers get an explicit ALL CLEAR. Alerts that simply stop breed alarm fatigue; a stand-down message builds the trust an early-warning service lives on.

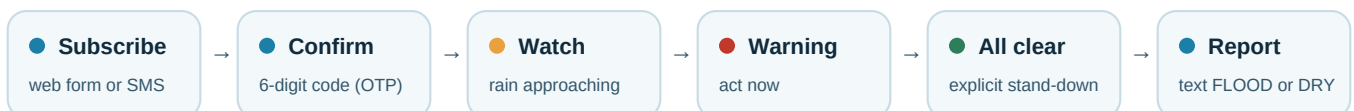
● Privacy by design (NDPR)

Explicit consent at sign-up, optional SMS-code confirmation so nobody can be subscribed (or moved) without holding the phone, PII visible only to authorized operators, and a 12-month data-retention policy.

● What it costs to run

The only marginal cost of the whole service is the SMS itself: NGN 5-15 (under 2 euro-cents) per message via Africa's Talking. Hosting, data and models run on free tiers - the service scales with need, not with budget.

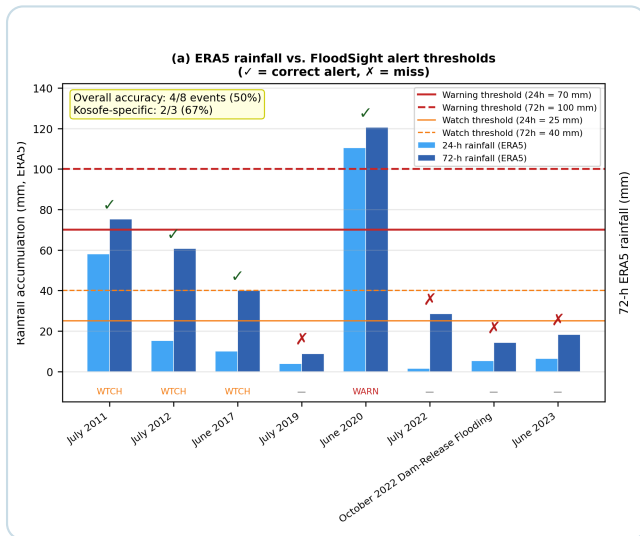
ONE SUBSCRIBER'S JOURNEY



WHERE THE PILOT STANDS TODAY

Live city-wide system · daily 06:00 SMS briefing · hourly alert dispatch · first resident cohort (8 subscribers) onboarded ahead of the 2026 rainy season · field-verification loop open and waiting for its first season of ground truth.

Validated against 13 years of real Lagos floods



Independent back-test: replaying 2011–2023 rainfall against FloodSight's thresholds. Across the full 18-event register (2011–2024) the system flags 14 of 18 documented Lagos floods (78%). The misses cluster in dam-release and data-artefact events — which is exactly what the roadmap addresses next.

Measurable results

- 14 / 18 historical Lagos floods (2011–2024) correctly flagged in back-testing
- Satellite-radar cross-check: SWMM drainage hotspots overlap observed 2024 flood extents
- 616 radar-labelled flood-depth observations from two real events anchor the ML layer
- Every alert, dispatch and field report is logged — precision & recall are recomputed after every event

A different kind of claim: radical honesty

Every training example is labelled observed vs. simulated, and the public API reports accuracy only on real observations — including where the model is still weak. In a field crowded with overstated “AI”, an early-warning system earns trust by publishing what it cannot yet do.

What the iF Social Impact Prize would fund

1 2026 rainy season

Full-season pilot across 15 LGAs: SMS credits for thousands of residents (≈ €0.12 per resident per season) and stipends for community verification reporters.

2 Close the misses

Integrate Oyan dam releases and NiHSA tide-gauge data — the two flood drivers rainfall alone cannot see — with agency partnerships.

3 Prove, then scale

Publish season precision/recall openly; retarget the one-config-file pipeline to a second West African coastal city with a local partner.

IN THE LANGUAGE OF THE FIVE iF CRITERIA

Solves a problem

— closes the last-mile warning gap for a flooding megacity

Moral-ethical standards

— consent, opt-out, privacy, calm language, honest metrics

Connection to design

— the product IS design: information design of one 160-character sentence

Effort vs. use value

— open data + free infrastructure → a city protected for the cost of its SMS

Positive experience

— two-way dialogue, an explicit all-clear, residents as partners not recipients